Human Step Movement Imitation For Training Privacy-Aware Intelligent Environments

Johannes Geyer[1], Matthias Melzer[2], Jan Dünnweber[2]
Ostbayerische Technische Hochschule Regensburg
Faculty of Computer Science and Mathematics
Regensburg, Germany
[1]firstName.lastName@st.oth-regensburg.de
[2]firstName.lastName@oth-regensburg.de

Amitrajit Sarkar
Ara Institute of Canterbury
Christchurch, New Zealand
amit.sarkar@ara.ac.nz

2026 22nd International Conference on Intelligent Environments (IE) | 979-8-3315-6450-6/26/$31.00 ©2026 IEEE | DOI: 10.1109/IE69249.2026.11539056

***Abstract*—Movement sensors are used, e.g., for automated lighting or for HVAC automation. Depending on the type of sensor (ultrasonic, radar, etc.), not only the presence but also the number of people in a building can be determined. ML models trained with movement data can support the interpretation of local movement patterns: *Situated intelligence* means putting elementary data, such as step sensor measurements, together and learning bottom-up which movement patterns may indicate processes such as crowd formation or hurried motion. Such dynamic inference mechanisms are useful in guidance systems for key public venues, including shopping malls or tourist attractions. Compared to video-based automation, situated AI is more adaptable and can also be used where filming is not allowed due to privacy. In this work, we discuss privacy-aware tracking for environments using floor sensors and training data we built synthetically, but realistically, by imitating human musculoskeletal biomechanics. For two feasibility case studies, one in a grocery store and one on a city bus, we explain how we track unidentified people and derive limited behavioral indicators only from foot movements in a narrow area that we covered with a sensor map.**

***Index Terms*—intelligent environments; smart retail; building automation; smart transportation; situated intelligence**

I. Introduction

Most people will agree that it is a good idea to align the schedules and routes of buses and the placement of new bus stops, taking into account the actual needs of passengers. However, for city councils, it is not trivial to determine where and when travelers would like to have a new stop. Therefore, the planning of new routes is typically based on map partitioning and demographic data such as the size of the working population in an area [1]. Crowding around the bus entrance or other dynamic data, such as average travel times, is generally ignored since this information is not acquired when only ticket sales data are collected. Therefore, no detailed passenger behavior analysis is conducted, although it could substantially improve the planning of bus routes [2].

Grocery store management strategies also depend on understanding customers' needs. If store operators knew what customers were likely to buy next, they could develop a more forward-looking and economically effective inventory management strategy. At the same time, they could run a more sustainable business that avoids food waste.

Using sales data for different goods and visitor data for different departments of a store, AI models can be trained to make useful predictions for sustainable inventory management strategies, which customers are likely to support [3].

However, most people will probably not agree with being filmed on a bus or in a store to create training data for an AI, regardless of the potential benefits to environmental protection or urban mobility. In particular, the recording of video material showing individuals and the digital post-processing of such material is generally refused or even prohibited by law in some places [4]. Therefore, tracking the behavior of people in public places like buses is typically done by conducting surveys on site [2]. One of the reasons why the results of such surveys are unreliable is that many relevant groups do not participate at all. These groups include people who are in a rush or people who are dissatisfied and could name weaknesses that need to be addressed. In statistics, this problem is known as the coverage error, which has already been studied for a long time with the aim of conducting more reliable surveys [5].

As an alternative to filming or collecting questionnaires, we describe an approach for detecting and characterizing local movement patterns without identifying individuals or intruding on their privacy. To overcome the problems with unavailable and/or sensitive data resulting from human behavior, we trained an AI model to perform a floor-sensor-based gait analysis. The footstep placement data which any person leaves unintentionally wherever she or he is walking are not data in need of protection. These data can be recorded in public places and processed electronically without the need for asking the originator. The data is verifiable, observed people remain anonymous and the local tracking data allows for deriving meaningful behavioral indicators, as explored in our experiments.

We raise the following three research questions (RQ):

RQ1: How reliably can Machine Learning (ML) detect footsteps from the data collected by floor sensor mats and assign the movement events to individuals?

RQ2: Can we generate useful synthetic data for training our model that recognizes people on sensor mats by simulating human biomechanics?

RQ3: What installation and operational factors influence the precision and reliability of such systems?

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

Concerning RQ1, we present a tracking pipeline that connects sensor mat measurements of a person's feet uniquely to a person using instance segmentation. Concerning RQ2, we describe our musculoskeletal simulation of human biomechanics and how we integrated it into standard software (Unity → Blender → Motion Capture) to generate step activation patterns. Regarding RQ3, we report on our case studies in a bus and in a shop. We report instance-level recognition metrics for our data samples and show how analyzing movement indicators, such as length of stay and peaks, can support planning-related insights.

For a comparison with computer-supported video surveillance, we describe a reference experiment on customer observation in a mall in New Zealand where public video recording is allowed [6]. In a follow-up experiment, we installed floor sensor mats in a German grocery store. In both experiments, the floor sensors provided more fine-grained local movement information where customers were counted. However, the customer movements could not be analyzed at the same level of detail. We conducted another experiment in a city bus, where we also installed our floor sensor mats. The recorded timing and movement data provided a clear understanding of where and when the bus was empty or overcrowded.

As a preparation procedure for the sensor experiments, we recorded ourselves walking across the sensor mats in the case study environment. By combining the recorded data with simulation data, we could train a reliable AI model, although the recorded movement data, originating from two of the authors, was limited and most input was generated by imitating human movements. Imitated human movements are also used for making avatars look realistic in interactive educational software or computer games [7]. It has also been shown that prompt reactions to human movements lead to safer interactions between people and robots [8]. However, step pattern imitation for privacy-aware local movement analysis in urban environments has not yet been comprehensively investigated elsewhere.

To address concerns regarding data collection and generation in relation to data protection and ethics, we emphasize that all data we collected are subject to the European Union's General Data Protection Regulation (GDPR) and New Zealand's Privacy Act 2020. All common legislative and ethical guidelines, including those of the participating universities, have been observed and complied with. Using the data, we perform a statistical analysis and explain what the measurements tell us about the observed movement patterns.

The remainder of this paper is structured as follows. In Section II, we discuss other work on interpreting behavioral processes based on event tracking. Section III shows how our neural network can interpret footstep events after we trained it with data that are synthetic but not random: using a biomechanical simulation to obtain realistic training data is the central methodological contribution of this work. In Section IV, we evaluate this idea in our experiments and compare the results from automated tracking with hand-annotated datasets.

II. Related Work

Process mining is a branch of data science related to the work we present, as it is also focused on behavioral data [9]. One problem of ERP systems, as they are often used in business analysis and economic process mining, is that the captured data is too coarse-grained for the computer-based detection of behavioral patterns. One reason for gaps in the data is that many systems retrieve data using *pull*-methods importing measurements on demand. Our floor sensors produce small data packets, which can be all *pushed* to our backend, leading to a more reliable measurement interpretation [10].

Alternatives to floor sensors that are sometimes used in warehouses for monitoring employees are shelf sensors and wearable devices, such as electronic wristbands [11]. While process mining can benefit from the combination of multiple data sources, maintaining situational awareness across an organization becomes more difficult with the number of systems involved [12].

Process mining systems for manufacturing, such as the ones described above, deal with processes that have an outcome allowing for the insertion of labels into the recorded timeline whenever a product is finished. Similarly, recorded dialog allows for the insertion of labels corresponding to the dialog topics [13]. Finding appropriate labels for footstep recordings at consecutive process states is more intricate. However, our work is not aimed at the reduction of processing steps or at understanding people's opinions concerning a topic, but rather at making beneficial adjustments to the environment which can be done without labeling the observed processes.

Other work on improving urban living via automation focuses on environmental characteristics and static conditions, such as the proportion of green space in an area, which can be captured electronically from photos without the need to track motions [14]. In another case study that used motion tracking, 397 test participants had to agree with the installation of a tracking app on their personal smartphones. The goal was ro predict when one of these persons would leave a store or visit the next one [15].

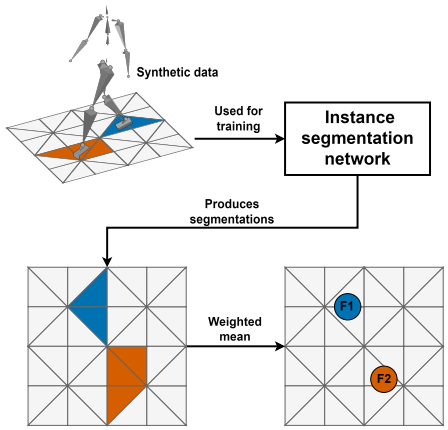


Fig. 1. The neural network is trained on synthetic data and produces instance masks. Masks are converted to feet positions using a weighted mean.

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

To track the behavior of groups of people, another case study combined smartphone-based trajectory identification with the use of *light detection and ranging* sensors (LiDAR) [16]. LiDAR sensors enable continuous data collection and may raise fewer privacy concerns than camera-based tracking, similar to our floor sensors. However, the combination of path analysis and LiDAR is quite complex: To assess crowd density, the authors had to combine the data from the smartphones' inertial sensors (accelerometers and gyroscopes) with measurements from multi-laser installations (15-20 LiDAR sensors). Other approaches to detecting and tracking people use WiFi beacons [17], Bluetooth Low Energy (BLE), or RFID beacons [18]. Similar to the worker-wristband study we discussed above, all participating persons must carry some kind of transponder device, which is not feasible outside industrial scenarios. These sensing modalities are relevant alternatives for privacy-aware tracking; however, an empirical comparison between footstep analysis and tracking people carrying transponders is beyond the scope of the case study presented in this work.

An AI model for anonymous foot localization and local tracking, as presented in this paper, was not used in the studies discussed above, which use various sensors and/or cameras in combination. We only use floor sensors and our AI model ignores identities, i.e., we focus on anonymous local movement patterns instead of recognizing who is being tracked, in contrast to other work that uses AI in gait analysis for recognizing individuals [19].

III. Methods: Precise people localization on floor sensing mats

Our method for detecting the position of people on floor sensor mats is based on the application of AI models from image processing to the data matrices of the floor sensor mats. The basic principle, in image processing and in our application as well, is the arrangement of the sensitive areas in a matrix of rectangles. This matrix is comparable to the pixels of a two-dimensional bitmap image. To further increase the trigger rate, the sensitive rectangular areas are divided alternately along the diagonals and anti-diagonals. This results in a special data map in which each coordinate pair corresponds to two data values: either a measurement at the bottom left and top right or a measurement at the bottom right and top left.

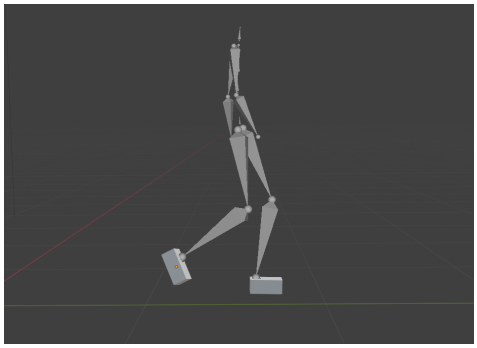


Fig. 2. The motion capture files are linked to cuboid meshes representing feet. Their intersections with the floor are used for generating synthetic sensor data.

The processing of data from adjacent patches works segment-by-segment, regardless of the shape of the single patches (or the sensor matrices they contain) which can be triangular (as in our case) or, for example, rhombic.

The only requirement is that all shapes are polygonal and do not overlap, as the neighborhood relationship must be clearly defined. The resulting data are used to train an *instance segmentation network* that can estimate the position of a person's feet walking across the floor sensors by analyzing the pattern generated on the sensor mat from multiple sensors. Using the segmentation masks generated by the network, a weighted average is calculated to estimate the positions of a person's feet, labeled F1 and F2 in Figure 1.

Regarding the design of the sensor mats, it should be noted that it is possible and common practice to adapt them to the floor layout without a loss of functionality [20]. In our experiments, we used the floor sensor system "SensFloor," which is primarily used in German hospitals for accident detection.

The following subsection III-A explains the process of generating synthetic data in more detail and justifies our choices of software technology. The implementation of the instance segmentation network and the algorithm for calculating the weighted mean value are discussed in more detail in the following subsection III-B.

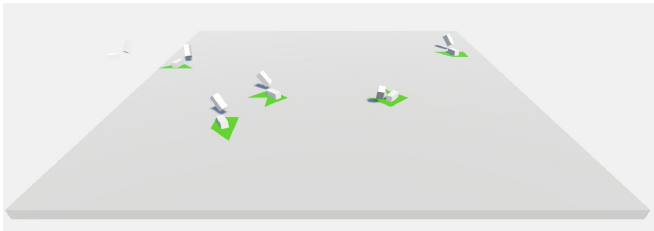


Fig. 3. Five simulated humans, with biomechanics modeled accordingly to motion capture files, create gait pattern data by walking in random directions.

A. Realistic Generation of Synthetic Data

In our synthetic data generation procedure, the setting of foot positions is based on a 3D simulation of walking humans triggering signals on simulated SensFloor patches. We modeled the former in *Unity*, due to the flexibility of this software with respect to input and output data[1]. Our patches are described using the so-called *Asset Bundles* in the *Unity Editor*. We divide 3D patch models into eight triangles per patch and export the position data to the simulation software *Blender*[2], where we run the movement simulation. Blender dynamically finds the regions where objects intersect, as required for locating the footsteps.

[1]Unity Technologies: Unity (2023), https://unity.com, game development platform

[2]Blender - a 3D modeling and rendering package. Blender Foundation, Stichting Blender Foundation, Amsterdam (2018), https://www.blender.org

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

Running simulations is also supported by popular *Google Sketchup* software, but only the *Pro Version* offers an *object intersection function* (under paid subscription [21]). In the free Blender software, where we process our topological data, we could also import *motion capture files*, which are publicly available and document various human motion styles, e.g., walking and running [22].

As an alternative to the recorded data, we also considered using a physics-based model, such as the *linear inverted pendulum* (LIP). LIP was originally developed to move biped robots [23] and allows for the introduction of cost functions and their optimization toward more energy-efficient movements [24], which is not relevant to us. One reason why many other studies on footstep simulation are based on physics is that regular movements are required. Recognizing the differences between these and human movements is helpful in identifying abnormalities that indicate pathological conditions [25], which is also not relevant to our work.

Comparisons between recorded and calculated movement data are common in medical big data analyses and often take into account, in addition to foot movements, detailed muscle action recordings, e.g., surface electromyography (EMG) [26]. Unlike a clinical study of human movements, our model requires only pixel-by-pixel masks as training data. The instance masks that represent our floor indicate pixel-by-pixel where sensor activations occur, similar to the description of moving objects, their edges, and contours in raster image processing. [27].

As shown in Figure 2, we link the motion capture data to cuboid meshes, which we use for to represent the feet of walking people - the reason for modeling "brick-shaped" feet is that Unity must store a mesh for every object used in Blender for calculating collision events. The patterns left by people walking over the sensors on the floor predominantly depend on the way their feet are placed relative to our triangular sensors. For example, a single foot can activate only the two top triangles of a patch when placed on the upper part. However, it can potentially activate up to eight triangles, which occurs when it is placed in the center of a patch. In the simulation, it is necessary to ensure a good variety of foot placements in relation to the sensors. Therefore, we randomly alter the starting position and orientation of the simulated humans every time we spawn another thread to simulate a new human instance in our walking simulation. An example simulation iteration can be seen in Figure 3, where five simulated humans walk across a grid of 10×10 floor patches and trigger different activation patterns.

Thus, our algorithm for creating simulated humans with their starting position and orientation performs the following steps: First, the number of simulated humans is chosen. This number varies between 1 and 8 for each simulation iteration. The algorithm then randomly assigns a starting position along any edge of the sensor area and an orientation between $[-60°, 60°]$ to each simulated human. The walking motion for each simulated human is provided through the motion capture file. When the paths of two simulated humans intersect and both activate the same cells on the sensor floor, one of the activations is ignored in our experiments. This simplification reduces the realism of heavily overlapping crowd situations and should be considered a limitation of the synthetic data generation.

We find that combining recorded measurements with a small amount of random noise produces suitable training data for an AI model capable of categorizing human movement events. This constitutes an additional result of our work.

B. Architecture and Implementation

To detect and differentiate between the feet of people walking on the floor sensors, we use instance segmentation with *pixel embeddings* and a subsequent *weighted average* calculation to estimate the position. These implementation decisions are justified for our applications since sensor floor mats are costly and, therefore, often used in small and strategically placed areas. For our supermarket experiments, we focus on the cash register area where two billboards are located. In the public bus, our sensor mats are placed close to the doors. Thus, we are processing dense data that are relevant for localized events such as waiting near a billboard or passenger movement near the bus doors. The embedding of pixels has shown clear advantages in such scenarios [28] over other AI models, e.g., a Support Vector Machine (SVM) that is suitable when a classification involves a large number of irrelevant features [29].

We embed the individual pixels into an n-dimensional space such that all pixels of one-foot instance are near each other and far away from the pixels of other instances. In a subsequent step, the pixel embeddings are clustered to assign them to an instance. We use a *discriminative loss* function and the *mean shift algorithm* for clustering, which have been proven to be useful for training a reliable model without recurrent mechanisms in computer vision [30].

Other instance segmentation networks typically have two stages: 1) the generation of region proposals; and 2) subimage segmentation [31]. In our case, region proposals are not reliable because the activation patterns produced by individual feet consist of a small number of pixels that are close to each other and often occlude.

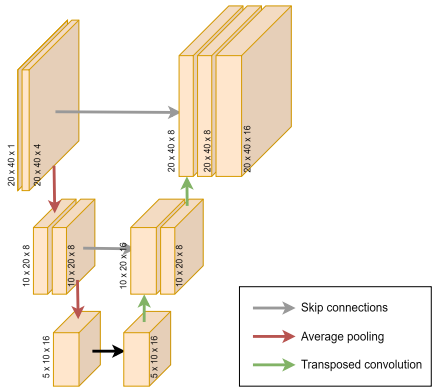
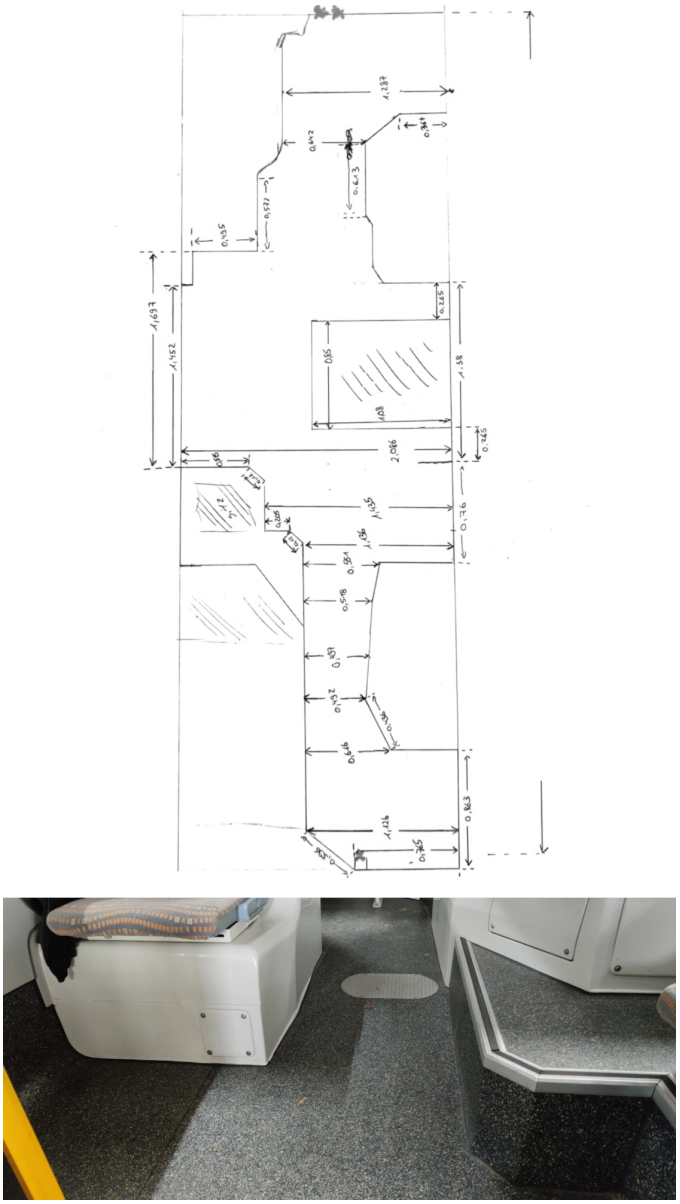


Fig. 4. Architecture of the instance segmentation network for detecting feet.

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.



(a) Permanent floor installation



(b) Layout and covered mats

Fig. 5. Our sensor mats in a city bus.

An advantage of the instance segmentation method with pixel embeddings, which we implemented, is that our AI model processes the image as a whole in one single segmentation stage and can better handle complex occlusions. It also does not need bounding boxes to provide additional information to differentiate between instances.

We work with a 10×10 grid of floor patches. Our neural network architecture is shown in Figure 4 and consists of three convolutional layers with a mean pooling between them on the descending branch. For the ascending branch, this structure is mirrored, and the pooling operation is replaced by transposed convolution for upsampling. In addition, two skip connections are used. For the pixel embedding, 16 dimensions are used in the last layer of the network. This network size is sufficient for our purposes. A larger network would be harder to train and more prone to overfitting. The embedding in the 16-dimensional space is done by calculating the loss. Instead of soft-max, as commonly used in instance segmentation, we use a discriminative loss function, since this loss function has shown better performance in detecting objects that are separated into multiple pieces, as required for our floor sensors: it pulls the embeddings towards a cluster center, while dissimilar data are pushed further away from each other. Our implementation of the model is based on U-Net [32] and was trained for 200 epochs on a training dataset with 25918 synthetically generated footstep samples.

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

The embeddings are provided by the network. In the next step, they are effectively grouped into instances using the mean shift algorithm. The mean-shift algorithm starts by selecting any pixel embedding as a cluster center and clusters all other pixel embeddings within a radius around it. In a second step, the mean of all clustered embeddings is calculated and chosen as the new center. All embeddings within a radius around this new center are clustered again. This process is repeated until the position converges and all cluster centers are in a dense region.

After the instance segmentations are obtained, the foot position is calculated using formula (1):

$$P = \frac{\sum_{i=1}^{n} w_i p_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

In this formula (which originates from the detection of accidents [33]) P is the position of the foot, w_i is the activation capacitance value, and p_i is the position of the centroid of the activated triangle. Calculating the position in this way is accurate since we take into account the different capacitance values. A position will always tend toward activations with higher values, i.e., where the greater part of the foot is placed.

We tested the detection algorithm on a synthetic validation dataset comprising 1541 entries and a hand-annotated dataset with 49 entries. In the synthetic validation dataset, the model scored a precision of 0.937 for an *intersection over union* (IoU) threshold of 0.95, while the precision in the hand-annotated dataset was slightly lower, as shown in Table I and Table II. Given the limited size of the hand-annotated dataset, these real-data results should be interpreted as an initial transferability check.

TABLE I
PRECISION AND RECALL METRICS OF THE NETWORK ON A SYNTHETIC VALIDATION DATASET.

IoU Threshold	**0.500**	**0.750**	**0.950**
Precision	0.973	0.941	0.937
Recall	0.973	0.941	0.937

TABLE II
PRECISION AND RECALL METRICS OF THE NETWORK ON A HAND-ANNOTATED REAL DATASET

IoU Threshold	**0.500**	**0.750**	**0.950**
Precision	0.989	0.887	0.887
Recall	0.989	0.887	0.887

IV. EXPERIMENTS

To motivate the two follow-up experiments showing the usefulness of our floor sensors, we start with a smart retail case study which we conducted in Christchurch, New Zealand, without using floor sensors.

Christchurch is a unique area for examining the relationship between pedestrian traffic and retail revenue. The city underwent a significant redevelopment after the 2010 and 2011 earthquakes, providing an opportunity to create modern infrastructure and amenities.

Data for our experiment were collected using surveillance cameras from stores representing six different categories: “apparel & personal goods”, “cafes, bars & restaurants”, “fuel & automotive”, “groceries & liquor”, “home & recreation” and “other retail”. The diagram in Figure 6 shows the mean daily count of pedestrians on the x-axis. The mean expenditure in New Zealand Dollars is shown on the y-axis. One might expect a linear relationship; that is, more customers produce more revenue. Obviously, such a relationship is *not* present in our data. Therefore, expenditures must be influenced by factors such as alternative shopping on-line or economic resilience.

In addition to synthetic validation (Table I), we evaluated detection quality using 49 manually annotated real images (Table II). The result was a precision/recall $= 0.887$ with an IoU of 0.95. This result suggests that our instance segmentation model can be transferred to real mats. However, given the limited size of the hand-annotated real dataset, this conclusion should be interpreted as preliminary. The video study in the mall serves only as contextual evidence of pedestrian buying behavior. It was *not* used to train or evaluate the foot detection model.

In the following two case studies, we not only count people but also explore movement-related indicators that may be associated with purchasing decisions and other local behavior patterns. These parameters probably include the range of available products and convenient and affordable access to shops, including parking and transportation options. Figure 5 shows that our installation covers the entry/exit area and allows us to observe people getting on and off the bus. In addition, we observe who takes a seat and how many passengers remain standing in the entry area until they get off. Figure 7 shows the results of the first experiment we conducted for a month with our sensor floor mat on a city bus in Regensburg.

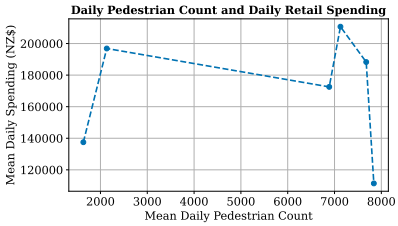


Fig. 6. Daily pedestrian count and daily retail expenditures

As shown in the right Figure 7, we observed peak times (left) and stepwise movements (right), which only occurred until there were fewer than 30 people on the bus. For this feasibility study, foot movement was classified as a step when two consecutive foot positions of a detected person were more than 30 cm apart. Likewise, we used more than 30 passengers as an operational definition of overcrowding. For data protection and operational reasons, we did not record synchronized videos during the trials in buses and supermarkets, as our experiments were designed as feasibility studies.

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

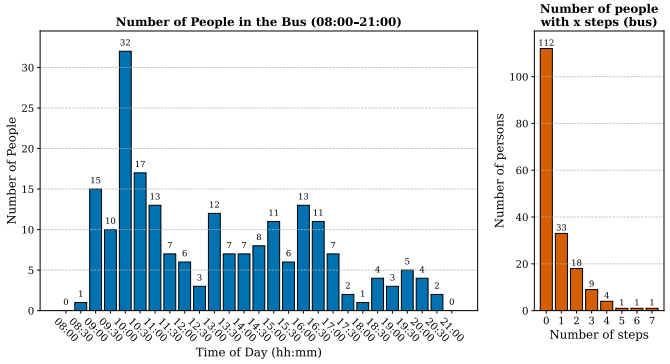


Fig. 7. Tracking results showing passenger behavior in a public city bus, Left: Peak time observation of density on the bus, Right: Step-wise movements while the bus was crowded

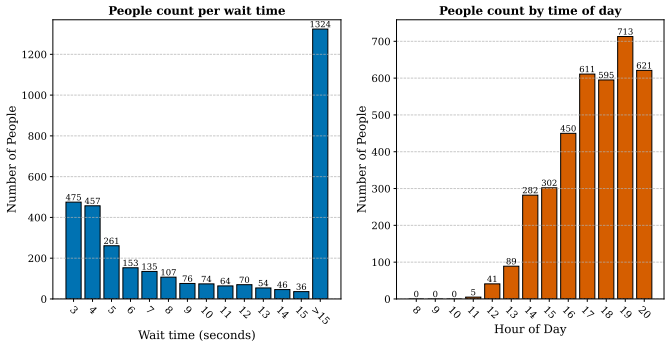


Fig. 8. Visitor distribution during business hours. Left: People grouped at a billboard, Right: Hourly count of customers.

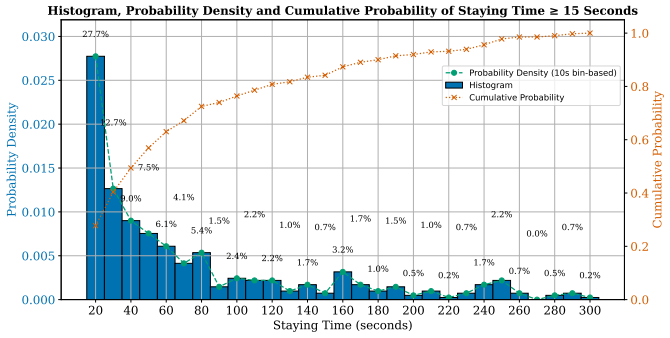


Fig. 9. Customer waiting times $\geq$ 15 s in 10-second bins, left y-axis: Probability Density, right y-axis: Cumulative Probability.

For the next experiment, we installed floor sensors in front of a digital billboard displaying new products at a grocery store in Regensburg, which we observed for one month. In our case study, we used heuristic grouping rules: whenever we observed people walking distances less than 70 cm within a given period, we treated this as a possible indicator of interest in the advertisements, as they stayed close to the billboard for a moment. In contrast, people leaving the sensor area after 3 seconds or less were treated as people who are probably not interested in the advertisements. Figure 8 left shows a fine-grained classification of the measurements: For durations between 3 and 15 seconds, we discretely counted the number of people staying in front of the digital billboard. The number of customers who stay in front of the billboard rapidly decreases after 5 or 6 seconds; that is, there are only a few customers whose attention was caught for a longer time. Less than 100 customers spent more than 10 seconds there, while more than 500 stayed for only 2 seconds. This is still more than the time required for a single step; i.e., all these customers should have noticed the billboard.

The result of a statistical analysis of the times people stood in front of the billboard is shown in Figure 9. We focused on people whom our AI recognizes as being in front of the billboard for more than 15 s. As a heuristic interpretation, those who spent this amount of time directly in front of the billboard are likely to have viewed the ad. Some false positives may remain; for example, people who stood near the billboard for longer periods for other reasons. This is underpinned by the approximately periodic length of stay corresponding to the duration of the advertisement running in a loop. In Figure 9, the value ranges are grouped into bins of 10 s. Many people turned away after 80 s, after 160 s, and around 240 s. These intervals correspond to the repetitions of the ads shown. These observations indicate that the AI can provide a plausible mapping between foot movements and people in this deployment.

The average customer count for one month of opening hours is shown in the right part of Figure 8 and was also verified using our on-site transcript. The highest customer density was observed during the evening, after many customers had finished their office hours. Duration measurements resulting from tracking customers' feet are another valuable outcome of our experiment. The simple counting of customers could also have been achieved by evaluating the cash registers.

Our observation of the staying durations clearly benefits from our AI model trained to understand foot movements. It provides additional insight into customer movement patterns, even when the store is crowded. Multiple values are taken into account: revenue, staying duration, customer count, and distances between feet (crowdedness). Thus, our data are also suitable for a *partial information decomposition* [34], which can classify values into unique, redundant, and synergistic ones.

Since, no additional video recording of the bus and the supermarket floor map deployments exist for the experiment time, our behavioral interpretations cannot be proven. However, our data interpretations are plausible, even from a more distant perspective, taking the diagrams shown in this section into account.

V. Conclusion

One research question studied in this work was how computers can analyze local movement patterns in public spaces when camera recordings are not available. Using floor sensor mats on a bus and in a supermarket, and training with biomechanically simulated data, we created an AI model for privacy-aware gait analysis. Our results suggest that ML can detect footsteps on floor sensor mats with promising precision,

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.

that synthetic biomechanics-based data are useful for training, and that sensor placement in small, strategically chosen areas strongly affects the usefulness of the measurements. While floor sensor mats are not suitable for fast large-area tracking, they offer a privacy-preserving option for localized planning support. Given the small annotated real dataset and the heuristic nature of the case studies without synchronized video ground truth, the behavioral findings should be understood as indicative; future work should extend the real-world validation and compare our floor patches to alternative privacy-aware people tracking approaches.

REFERENCES

[1] T. Stadler et al., "A Method for the Optimized Placement of Bus Stops Based on Voronoi Diagrams," in *Proceedings of the 55th Hawaii International Conference on System Sciences (HICSS55)*, 2022. DOI: 10.24251/HICSS.2022.694.

[2] T. Stadler et al., "Bus Demand Forecasting for Rural Areas Using XGBoost and Random Forest Algorithm," en, in *Computer Information Systems and Industrial Management*, K. Saeed et al., Eds., Cham: Springer International Publishing, 2021. DOI: 10.1007/978-3-030-84340-3_36.

[3] W. Huang et al., "Optimizing smart retail by experiment using an online AI model exploration interface," in *Proceedings of the winter simulation conference*, Orlando, Florida, USA: to appear in IEEE Press, 2024. DOI: 10.1109/WSC63780.2024.10838897.

[4] E. Ullmann-Margalit, "The case of the camera in the kitchen: Surveillance, privacy, sanctions, and governance," en, *Regulation & Governance*, 2008. DOI: 10.1111/j.1748-5991.2008.00047.x.

[5] F. Szuster et al., "Nonsampling Error in Surveys.," *The Statistician*, 1994. DOI: 10.2307/2348357.

[6] Geoffrey Palmer et al., *Right of Privacy, NZLC SP19, New Zealand Law Commission, Wellington, ISBN 978-1-877316-43-2*, 2008.

[7] A. Ikkala et al., "Breathing Life Into Biomechanical User Models," en, in *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, Bend OR USA: ACM, 2022. DOI: 10.1145/3526113.3545689.

[8] F. Michaud et al., "Pushing a robot along - a natural interface for human-robot interaction," in *2010 IEEE International Conference on Robotics and Automation*, 2010. DOI: 10.1109/ROBOT.2010.5509584.

[9] A. Weijters et al., "Rediscovering workflow models from event-based data using little thumb," *Integrated Computer-Aided Engineering*, vol. 10, no. 2, 2003. DOI: 10.3233/ICA-2003-10205.

[10] P. Girao et al., "Remembering old practices to validate measurement data," *IEEE Instrum. Meas. Mag.*, 2022. DOI: 10.1109/MIM.2022.9908264.

[11] Y. Gong et al., "Field validation of beacon-based indoor tracking and localization system for construction workers," *KSCE Journal of Civil Engineering*, vol. 29, no. 2, 2025. DOI: 10.1016/j.kscej.2024.100017.

[12] A. Orebaugh et al., "Information security continuous monitoring for federal information systems and organizations," National Institute of Standards and Technology (NIST), Washington, DC, Special Publication 800-137, 2011. DOI: 10.6028/NIST.SP.800-137.

[13] A. Alves et al., "Unhide: A novel framework for unsupervised human-interpretable dialogue exploration," *IEEE Access*, 2025. DOI: 10.1109/ACCESS.2025.3632654.

[14] C. Benjamin et al., "Urban greening, property and more-than-human commoning," *Australian Geographer*, 2020. DOI: 10.1080/00049182.2019.1655828.

[15] Y. Hahm et al., "The effect of built environments on the walking and shopping behaviors of pedestrians; a study with gps experiment in sinchon retail district in seoul, south korea," *Cities*, 2019. DOI: 10.1016/j.cities.2019.01.020.

[16] H. Yamaguchi et al., "A human tracking and sensing platform for enabling smart city applications," en, in *Proceedings of the Workshop Program of the 19th International Conference on Distributed Computing and Networking*, New York, NY, USA: ACM, Jan. 2018. DOI: 10.1145/3170521.3170534.

[17] K. Gebru, "A Privacy-preserving Scheme for Passive Monitoring of People's Flows through WiFi Beacons," in *2022 IEEE 19th Annual Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA: IEEE, 2022. DOI: 10.1109/CCNC49033.2022.9700591.

[18] E. Cay et al., "Beacons for indoor positioning," in *2017 International Conference on Engineering and Technology (ICET)*, Antalya: IEEE, 2017. DOI: 10.1109/ICEngTechnol.2017.8308143.

[19] B. Zhou et al., "The carpet knows: Identifying people in a smart environment from a single step," in *2017 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)*, 2017. DOI: 10.1109/PERCOMW.2017.7917618.

[20] P. Srinivasan et al., "A pressure sensing floor for interactive media applications," in *Proceedings of the 2005 ACM SIGCHI International Conference on Advances in Computer Entertainment Technology*, ser. ACE '05, Valencia, Spain: Association for Computing Machinery, 2005. DOI: 10.1145/1178477.1178526.

[21] K. Murdock, *Google SketchUp and SketchUp Pro 7 bible*. Indianapolis: John Wiley & Sons, 2009, ISBN: 978-0-470-29229-7.

[22] M. Kobayashi et al., *Motion Capture Dataset for Practical Use of AI-based Motion Editing and Stylization*, 2023. DOI: 10.48550/arXiv.2306.08861.

[23] S. Kajita et al., "Study of dynamic biped locomotion on rugged terrain-theory and basic experiment," in *Fifth International Conference on Advanced Robotics 'Robots in Unstructured Environments*, 1991. DOI: 10.1109/ICAR.1991.240688.

[24] O. Urbann et al., "Flexible linear inverted pendulum model for cost-effective biped robots," in *2015 IEEE-RAS 15th International Conference on Humanoid Robots*, 2015. DOI: 10.1109/HUMANOIDS.2015.7363525.

[25] J. Higginson et al., "Muscle contributions to support during gait in an individual with post-stroke hemiparesis," *Journal of Biomechanics*, 2006. DOI: 10.1016/j.jbiomech.2005.05.032.

[26] F. Di Nardo et al., "Machine learning for detection of muscular activity from surface emg signals," *Sensors*, 2022. DOI: 10.3390/s22093393.

[27] J. Lazarow et al., "Instance segmentation with mask-supervised polygonal boundary transformers," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. DOI: 10.1109/CVPR52688.2022.00434.

[28] Y. Wu et al., "Improving pixel embedding learning through intermediate distance regression supervision for instance segmentation," in *Computer Vision – ECCV 2020 Workshops*, A. Bartoli et al., Eds., Springer International Publishing, 2020. DOI: 10.1007/978-3-030-65414-6_16.

[29] A. Landeros et al., "Algorithms for sparse support vector machines," *Journal of Computational and Graphical Statistics*, 2023. DOI: 10.1080/10618600.2022.2146697.

[30] B. De Brabandere et al., "Semantic Instance Segmentation for Autonomous Driving," in *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Honolulu, HI: IEEE, 2017. DOI: 10.1109/CVPRW.2017.66.

[31] J. M. Molina et al., "Advances in instance segmentation: Technologies, metrics and applications in computer vision," *Neurocomputing*, vol. 625, 2025. DOI: 10.1016/j.neucom.2025.129584.

[32] O. Ronneberger et al., "U-Net: Convolutional Networks for Biomedical Image Segmentation," en, in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, N. Navab et al., Eds., Series Title: Lecture Notes in Computer Science, Cham: Springer International Publishing, 2015. DOI: 10.1007/978-3-319-24574-4_28.

[33] R. Bagarotti et al., "An Algorithm for Estimating Gait Parameters Through a Commercial Sensorized Carpet," in *2018 IEEE 4th International Forum on Research and Technology for Society and Industry (RTSI)*, Palermo: IEEE, 2018. DOI: 10.1109/RTSI.2018.8548404.

[34] S. Parkinson et al., "Characterizing high-order interactions between eye movement and head motion variables in augmented reality-based navigation experience," in *2025 IEEE International Conference on Multimedia and Expo (ICME)*, 2025. DOI: 10.1109/ICME59968.2025.11209820.

Authorized licensed use limited to: Ostbayerische Technische Hochschule Regensburg. Downloaded on September 25,2026 at 09:21:56 UTC from IEEE Xplore. Restrictions apply.